

Galois-theoretic classification of Kochen–Specker sets in dimension three

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Abstract

We classify the two-element coordinate alphabets $\mathcal{A} = \{0, \pm 1, \pm x\}$ with x an algebraic integer in a quadratic extension $K/\mathbb{Q}$ (or $x = \pm 2$, the rational case) for which the *raw* ray set $S(\mathcal{A}) \subset \mathbb{C}^3$, before cross-product completion, is KS-uncolorable. The classification gives three alphabet-invariant conditions, each controlled by the Galois norm and trace of x :

- (R) $x = \pm 2$ (*rational*),
- (i) $\mathrm{Tr}_{K/\mathbb{Q}}(x) = 0$ and $|x|^2 = 2$ (*pure modulus-2*),
- (ii) $N_{K/\mathbb{Q}}(x) = 1$ and $|\mathrm{Tr}_{K/\mathbb{Q}}(x)| = 1$ (*phase*).

Sufficiency is witnessed by explicit KS constructions (CK-31 for (R), Peres-33 and the $\mathbb{Z}[\sqrt{-2}]$ -33 set for (i), the Eisenstein-33 set for (ii)). Necessity is proved by a finite enumeration of three-term vanishing sums over the product set $\bar{\mathcal{A}} \cdot \mathcal{A}$, combined with SMT verification of the resulting colorability obstruction, showing that when no condition holds the orthogonality graph admits an explicit $\{0, 1\}$ -coloring. As a corollary we obtain a norm-and-trace characterization of the raw KS-contextual quadratic alphabets. A parallel survey under cross-product completion [3] identifies three further algebraic islands (Gaussian, Heegner-7, golden ratio) that fall outside the raw classification; establishing a Galois-theoretic classification of the completed six-island picture remains an open problem. The classification connects to the result of Cortez, Morales, and Reyes that $\mathbb{Z}[1/6]$ is the minimal ring supporting KS contextuality in dimension 3.

1 Introduction

The Kochen–Specker (KS) theorem [1] establishes that quantum measurements in Hilbert spaces of dimension $d \geq 3$ cannot be explained by noncontextual hidden-variable theories. The proof proceeds by exhibiting a finite set of rays—a *KS set*—that admits no consistent $\{0, 1\}$ -valued coloring respecting the rules: (I) orthogonal rays cannot both receive the value 1, and (II) every orthogonal triple (triad) contains exactly one 1. The existence of such a set refutes the assumption that measurement outcomes pre-exist independently of which other compatible measurements are performed alongside them.

The starting observation behind the present work is visible in the cube diagrams of Peres’s textbook [4]: two of the three known three-dimensional KS constructions—the 31-vector Conway–Kochen set (CK-31) and the 33-vector Peres set—are displayed side by side, yet their coordinate arithmetics differ in a suggestive way. CK-31 uses the integer alphabet $\{0, \pm 1, \pm 2\}$, with every vertex labelled by 1s and 2s. The Peres set uses $\{0, \pm 1, \pm \sqrt{2}\}$, replacing the “2” with “ $\sqrt{2}$.” Both share the same two-element template $\{0, \pm 1, \pm x\}$; only the generator x changes. This pattern invites a natural question: *what happens if we vary x systematically?* Could coordinates built from $1/\sqrt{2}$, or $\sqrt{3}$, or a cube root of 2, or a sixth root of unity produce new—perhaps smaller—KS sets?

A computational survey [3] pursued this question across quadratic, cyclotomic, and golden-ratio number fields, fixing a two-element coordinate alphabet $\mathcal{A} = \{0, \pm 1, \pm x\}$ with x a ring-of-integers generator, generating the induced ray set, and testing for KS-uncolorable subsets. The survey found six distinct algebraic *islands*:

- $\mathbb{Z}$: integer alphabet, CK-31 (31 vectors);
- $\mathbb{Z}[\sqrt{2}]$: Peres-type set (33 vectors);
- $\mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$ (Eisenstein integers): Cabello-type set (33 vectors);
- $\mathbb{Z}[\sqrt{-2}]$: 33 vectors;
- $\mathbb{Z}\left[\frac{1+\sqrt{-7}}{2}\right]$ (Heegner-7 ring): 43 vectors;
- $\mathbb{Q}(\varphi)$, $\varphi = \frac{1+\sqrt{5}}{2}$ (golden-ratio field): 52 vectors.

The survey identified two recurring mechanisms driving KS-uncolorability: a *modulus-2* mechanism ($|x|^2 = 2$ in the first three cases above) and a *phase* mechanism ($1 + \omega + \omega^2 = 0$ in the Eisenstein case, a degenerate instance of a unit vanishing sum). These patterns were, however, established computationally and without a unifying algebraic explanation.

This paper closes that gap for the family of quadratic extensions. We prove that the two mechanisms are the *only* ones capable of producing KS-uncolorable *raw* ray sets from a two-element quadratic alphabet, and that each is characterized precisely by the Galois norm and trace of the ring generator. The proof classifies three-term vanishing sums over the product alphabet $\bar{\mathcal{A}} \cdot \mathcal{A}$, determines the resulting orthogonality-graph structure, and shows that when neither mechanism is active the constraint graph decomposes into mutually independent subproblems each admitting an explicit valid coloring.

Theorem 1. *Let $\mathcal{A} = \{0, \pm 1, \pm x\}$ with $x \notin \{0, \pm 1\}$ an algebraic integer in a quadratic extension $K = \mathbb{Q}(\sqrt{d})/\mathbb{Q}$, or $x = \pm 2$ (the degenerate rational case $K = \mathbb{Q}$). The raw ray set $S(\mathcal{A}) \subset \mathbb{C}^3$ is KS-uncolorable if and only if one of:*

- (R) $x = \pm 2$ (rational degenerate case);
- (i) $\text{Tr}_{K/\mathbb{Q}}(x) = 0$ and $|x|^2 = 2$ (pure modulus-2: $x = \pm\sqrt{2}$ or $x = \pm\sqrt{-2}$);
- (ii) $N_{K/\mathbb{Q}}(x) = 1$ and $|\text{Tr}_{K/\mathbb{Q}}(x)| = 1$ (phase, i.e. $1 + x + \bar{x} = 0$ or $1 - x - \bar{x} = 0$).

Here $N_{K/\mathbb{Q}}(x)$ and $\text{Tr}_{K/\mathbb{Q}}(x)$ denote the Galois norm and trace of x (Section 2). The three conditions are each invariant under $x \mapsto -x$ and under $x \mapsto \bar{x}$, hence depend only on the alphabet $\mathcal{A}$ itself.

Remark 1 (Raw vs. completed). *Theorem 1 classifies the raw ray set $S(\mathcal{A})$, before cross-product completion. A systematic computational survey [3] identifies six algebraic islands in dimension three once the ray pool is closed under cross products: the three captured above, together with the Gaussian alphabet ($x = 1 + i$), the Heegner-7 alphabet ($x = (1 + \sqrt{-7})/2$), and the golden-ratio alphabet ($x = \varphi$). The latter three are colorable at the raw level (insufficient triad density) and become KS-uncolorable only after completion introduces additional rays. Classifying the completed six-island picture by Galois-theoretic invariants remains an open problem.*

The proof occupies Section 3. Section 4 derives corollaries including the imaginary quadratic classification and the Galois symmetry of the orthogonality graph. Section 5 discusses connections to the Cortez–Morales–Reyes ring-theoretic result.

2 Setup

2.1 Quadratic extensions and their arithmetic

Let $d \in \mathbb{Z}$ be squarefree and $K = \mathbb{Q}(\sqrt{d})$ the associated quadratic extension. The ring of integers $\mathcal{O}_K$ has the form $\mathbb{Z}[x]$ where the generator x is:

$$x = \begin{cases} \sqrt{d} & \text{if } d \equiv 2, 3 \pmod{4}, \\ \frac{1 + \sqrt{d}}{2} & \text{if } d \equiv 1 \pmod{4}. \end{cases}$$

The Galois group $\text{Gal}(K/\mathbb{Q}) = \{1, \sigma\}$ is generated by the non-trivial automorphism $\sigma(\sqrt{d}) = -\sqrt{d}$. From this we define the *Galois norm* and *Galois trace* of any $\alpha \in K$:

$$N_{K/\mathbb{Q}}(\alpha) = \alpha \cdot \sigma(\alpha), \quad \text{Tr}_{K/\mathbb{Q}}(\alpha) = \alpha + \sigma(\alpha).$$

For the generator x these take concrete values depending on d . In the case $d \equiv 2, 3 \pmod{4}$: $N_{K/\mathbb{Q}}(x) = \sqrt{d} \cdot (-\sqrt{d}) = -d$ and $\text{Tr}_{K/\mathbb{Q}}(x) = 0$. In the case $d \equiv 1 \pmod{4}$: $N_{K/\mathbb{Q}}(x) = \frac{1+\sqrt{d}}{2} \cdot \frac{1-\sqrt{d}}{2} = \frac{1-d}{4}$ and $\text{Tr}_{K/\mathbb{Q}}(x) = 1$.

2.2 The coordinate alphabet and ray set

Fix a quadratic extension $K/\mathbb{Q}$ with generator x as above. Define the *two-element coordinate alphabet*

$$\mathcal{A} = \{0, +1, -1, +x, -x\}.$$

The *ray set* $S(\mathcal{A})$ consists of all projectively distinct nonzero vectors $v \in \mathcal{A}^3 \setminus \{0\}$, where two vectors v, w represent the same ray if $v = \lambda w$ for some scalar $\lambda \in \mathbb{C}^\times$. Orthogonality is defined via the standard Hermitian inner product:

$$\langle v | w \rangle = \sum_{k=1}^3 \bar{v}_k w_k = 0.$$

2.3 The product set and vanishing sums

For the orthogonality analysis the key object is the *product set*

$$\bar{\mathcal{A}} \cdot \mathcal{A} = \{\bar{a}b : a, b \in \mathcal{A}\} = \{0, \pm 1, \pm x, \pm \bar{x}, \pm |x|^2\}.$$

The inner product $\langle v|w \rangle$ for $v, w \in \mathcal{A}^3$ is a sum of at most three terms, each drawn from $\bar{\mathcal{A}} \cdot \mathcal{A}$. Orthogonality $\langle v|w \rangle = 0$ therefore reduces to a *vanishing sum* of one, two, or three elements from this product set.

A *primitive three-term vanishing sum* from $\bar{\mathcal{A}} \cdot \mathcal{A}$ is an identity $\alpha + \beta + \gamma = 0$ with $\alpha, \beta, \gamma \in \bar{\mathcal{A}} \cdot \mathcal{A}$, not reducible to a shorter vanishing sum. The classification of such sums—there are finitely many modulo sign and permutation—drives the proof of Theorem 1.

Explicitly, the product set elements are controlled by two invariants of the extension: For imaginary quadratic fields, complex conjugation coincides with σ on K , so $|x|^2 = x\bar{x} = |N_{K/\mathbb{Q}}(x)|$ and $x + \bar{x} = \text{Tr}_{K/\mathbb{Q}}(x)$; for real quadratic fields, $\bar{x} = x$ and $|x|^2 = x^2$. Condition (i) $|x|^2 = 2$ means the product set contains an element of absolute value 2, enabling cancellations $1 + 1 - |x|^2 = 0$. Condition (ii) means $1 + x + \bar{x} = 1 + \text{Tr}_{K/\mathbb{Q}}(x) = 0$, a unit-modulus vanishing sum as in $1 + \omega + \omega^2 = 0$.

3 Main theorem

We restate the main result for convenience.

Theorem 2 (Galois classification, restated). *Let $\mathcal{A} = \{0, \pm 1, \pm x\}$ with $x \notin \{0, \pm 1\}$ an algebraic integer in a quadratic extension $K = \mathbb{Q}(\sqrt{d})/\mathbb{Q}$, or $x = \pm 2$ (rational case). The raw ray set $S(\mathcal{A}) \subset \mathbb{C}^3$ is KS-uncolorable if and only if one of:*

- (R) $x = \pm 2$ (rational);
- (i) $\text{Tr}_{K/\mathbb{Q}}(x) = 0$ and $|x|^2 = 2$ (pure modulus-2);
- (ii) $N_{K/\mathbb{Q}}(x) = 1$ and $|\text{Tr}_{K/\mathbb{Q}}(x)| = 1$ (phase).

Sufficiency. Each condition is witnessed by an explicit KS construction.

- Condition (R) $x = \pm 2$ gives the 31-vector Conway–Kochen set CK-31 (rigidity proved by Trandafir–Cabello [2]).
- Condition (i) is realized uniquely (up to sign) by $x = \pm\sqrt{2}$ and $x = \pm\sqrt{-2}$. The real case gives the 33-vector Peres set [5]; the imaginary case gives the 33-vector $\mathbb{Z}[\sqrt{-2}]$ set [3], which is graph-isomorphic to Peres.
- Condition (ii) is realized by $x = \omega = e^{2\pi i/3} = (-1 + \sqrt{-3})/2$ ($N_{K/\mathbb{Q}}(\omega) = 1$, $\text{Tr}_{K/\mathbb{Q}}(\omega) = -1$), yielding the 33-vector Eisenstein set [6]; equivalently, by the canonical generator $x = (1 + \sqrt{-3})/2$ ($N_{K/\mathbb{Q}}(x) = 1$, $\text{Tr}_{K/\mathbb{Q}}(x) = +1$), which produces the same alphabet $\{0, \pm 1, \pm x\} = \{0, \pm 1, \pm \bar{\omega}\}$.

Other alphabets with $|x|^2 = 2$ but $\text{Tr}_{K/\mathbb{Q}}(x) \neq 0$ (the Gaussian generator $x = 1 + i$ and the Heegner-7 generator $x = (1 + \sqrt{-7})/2$) produce ray pools with orthogonal pairs but *insufficient triad density* to force uncolorability at the raw level (Lemma 2, Step 1).

Known KS constructions over these alphabets [3] require cross-product completion and lie outside the scope of the present raw classification; see Remark 1. $\square$

Necessity. We prove the contrapositive: if neither (i) nor (ii) holds, then $S(\mathcal{A})$ admits a valid $\{0, 1\}$ -coloring. The argument proceeds through two lemmas.

Lemma 1 (Vanishing sum enumeration). *Let $x \notin \{0, \pm 1\}$ be a ring-of-integers generator with conjugate $\bar{x} = \sigma(x)$. The primitive three-term vanishing sums $t_1 + t_2 + t_3 = 0$ with each $t_k \in \{0, \pm 1, \pm x, \pm \bar{x}, \pm |x|^2\}$ are, up to sign and permutation, exactly those listed in Table 1. In particular, every such sum constrains $(N, T) = (N_{K/\mathbb{Q}}(x), \text{Tr}_{K/\mathbb{Q}}(x))$ to one of the rows in the table.*

Table 1: Primitive three-term vanishing sums from the product set $\bar{\mathcal{A}} \cdot \mathcal{A}$. Here $N = N_{K/\mathbb{Q}}(x)$ and $T = \text{Tr}_{K/\mathbb{Q}}(x)$.

Pattern	Constraint	N, T
$1 + 1 + (-x) = 0$	$x = 2$	(rational)
$1 + 1 + (- x ^2) = 0$	$ x ^2 = 2$	± 2 , varies
$1 + x + \bar{x} = 0$	$\text{Tr}_{K/\mathbb{Q}}(x) = -1$	varies, -1
$1 - x - \bar{x} = 0$	$\text{Tr}_{K/\mathbb{Q}}(x) = 1$	varies, 1
$1 + x + (- x ^2) = 0$	$ x ^2 = 1 + x$	$-1, 1$ (golden)
$ x ^2 + x ^2 + (-1) = 0$	$ x ^2 = \frac{1}{2}$	$\equiv$ row 2
$x + x + (- x ^2) = 0$	$ x ^2 = 2x$	$\equiv$ row 1

Proof. The product set $\bar{\mathcal{A}} \cdot \mathcal{A}$ consists (up to sign) of $\{0, 1, x, \bar{x}, |x|^2\}$. A primitive three-term vanishing sum $\alpha + \beta + \gamma = 0$ with each term drawn from $\pm\{1, x, \bar{x}, |x|^2\}$ must satisfy two independent constraints: the *magnitude* constraint (the three terms must form a closed triangle in $\mathbb{C}$) and the *algebraic* constraint (the sum must vanish exactly in the ring $\mathbb{Z}[x, \bar{x}]$).

Writing each term as $\pm s_k$ with $s_k \in \{1, x, \bar{x}, |x|^2\}$, the vanishing sum becomes a polynomial identity in x and $\bar{x}$. Since $\{1, x\}$ is a $\mathbb{Q}$ -basis of K and $\bar{x} = T - x$ (where $T = \text{Tr}_{K/\mathbb{Q}}(x)$), every product-set element has a unique representation $a + bx$ with $a, b \in \mathbb{Q}$. The representations depend on whether K is real or imaginary quadratic.

Imaginary case ($d < 0$): complex conjugation coincides with σ on K , so $|x|^2 = x\bar{x} = N_{K/\mathbb{Q}}(x)$:

$$1 \mapsto (1, 0), \quad x \mapsto (0, 1), \quad \bar{x} \mapsto (T, -1), \quad |x|^2 \mapsto (N, 0).$$

Real case ($d > 0$): x is real, so $\bar{x} = x$ in the product set (the entries $\pm \bar{x}$ collapse to $\pm x$), and $|x|^2 = x^2 = Tx - N$ via the minimal polynomial $x^2 - Tx + N = 0$:

$$1 \mapsto (1, 0), \quad x \mapsto (0, 1), \quad \bar{x} = x \mapsto (0, 1), \quad |x|^2 = x^2 \mapsto (-N, T).$$

In the enumeration below, rows involving $\bar{x}$ as distinct from x arise only in the imaginary case. Rows involving $|x|^2$ are labeled (I) for imaginary and (R) for real where the constraints differ.

A vanishing sum $\alpha + \beta + \gamma = 0$ with each $\alpha, \beta, \gamma \in \pm\{1, x, \bar{x}, |x|^2\}$ yields two rational equations: the coefficient of 1 vanishes and the coefficient of x vanishes. The four unsigned elements $\{1, x, \bar{x}, |x|^2\}$ give exactly $\binom{4+2}{3} = 20$ unordered triples with repetition. We enumerate all 20, with all sign assignments $(\varepsilon_1, \varepsilon_2, \varepsilon_3) \in \{\pm 1\}^3$ modulo the overall sign and permutation symmetry:

Triple	Coeff. of 1	Coeff. of x	Solution
<i>Triples from $\{1\}$ only:</i>			
$\{1, 1, 1\}$	$\pm 1 \pm 1 \pm 1 = 0$	$0 = 0$	impossible
<i>Triples from $\{1, x\}$:</i>			
$\{1, 1, x\}$	$\pm 1 \pm 1 = 0$	$\pm 1 = 0$	impossible
$\{1, x, x\}$	$\pm 1 = 0$	$\pm 1 \pm 1 = 0$	impossible
$\{x, x, x\}$	$0 = 0$	$\pm 1 \pm 1 \pm 1 = 0$	impossible
<i>Triples involving $\bar{x}$ (imaginary only, since $\bar{x} = x$ for real):</i>			
$\{1, 1, \bar{x}\}$	$\pm 1 \pm 1 \pm T = 0$	$\mp 1 = 0$	impossible
$\{1, x, \bar{x}\}$	$1 + T = 0$	$1 - 1 = 0$	$T = -1$ (Row 3a)
$\{1, x, \bar{x}\}$	$1 - T = 0$	$-1 + 1 = 0$	$T = 1$ (Row 3b)
$\{1, \bar{x}, \bar{x}\}$	$\varepsilon_1 + (\varepsilon_2 + \varepsilon_3)T = 0$	$-(\varepsilon_2 + \varepsilon_3) = 0$	impossible ($\varepsilon_1 = 0$)
$\{x, x, \bar{x}\}$	$T = 0$	$\pm 2 \mp 1 = 0$	impossible
$\{x, \bar{x}, \bar{x}\}$	$(\varepsilon_2 + \varepsilon_3)T = 0$	$\varepsilon_1 - (\varepsilon_2 + \varepsilon_3) = 0$	impossible [‡]
$\{\bar{x}, \bar{x}, \bar{x}\}$	$(\sum \varepsilon_k)T = 0$	$-\sum \varepsilon_k = 0$	impossible ($\sum \varepsilon_k \neq 0$)
<i>Triples involving $x ^2$ — split (I) = imaginary, (R) = real:</i>			
$\{1, 1, x ^2\}$ (I)	$1 + 1 - N = 0$	$0 = 0$	$N = 2$ (Row 2)
$\{1, 1, x ^2\}$ (R)	$1 + 1 + N = 0$	$-T = 0$	$N = -2, T = 0$ (Row 2)
$\{1, x, x ^2\}$ (I)	$1 - N = 0$	$1 = 0$	impossible
$\{1, x, x ^2\}$ (R)	$1 + N = 0$	$1 - T = 0$	$N = -1, T = 1$ (Row 4)
$\{1, \bar{x}, x ^2\}$ (I)	$1 + T - N = 0$	$-1 = 0$	impossible
$\{x, x, x ^2\}$ (I)	$\varepsilon_3 N = 0$	$\varepsilon_1 + \varepsilon_2 = 0$	impossible ($N \neq 0$)
$\{x, x, x ^2\}$ (R)	$-\varepsilon_3 N = 0$	$\varepsilon_1 + \varepsilon_2 + \varepsilon_3 T = 0$	impossible ($N \neq 0$)
$\{x, \bar{x}, x ^2\}$ (I)	$T - N = 0$	$0 = 0$	$N = T$ †
$\{\bar{x}, \bar{x}, x ^2\}$ (I)	$2T - N = 0$	$-2 = 0$	impossible
$\{\bar{x}, x ^2, x ^2\}$ (I)	$\varepsilon_1 T + (\varepsilon_2 + \varepsilon_3)N = 0$	$-\varepsilon_1 = 0$	impossible
$\{ x ^2, x ^2, 1\}$ (I)	$2N - 1 = 0$	$0 = 0$	$N = 1/2$ ($\equiv$ Row 2)
$\{ x ^2, x ^2, 1\}$ (R)	$-2N + 1 = 0$	$2T = 0$	$N = 1/2, T = 0$ ($\equiv$ Row 2)
$\{ x ^2, x ^2, x\}$ (I)	$2N = 0$	$1 = 0$	impossible
$\{ x ^2, x ^2, x\}$ (R)	$-2N = 0$	$2T + 1 = 0$	impossible ($N \neq 0$)
$\{ x ^2, x ^2, x ^2\}$ (I)	$\pm N \pm N \pm N = 0$	$0 = 0$	$N = 0$ (excluded)

[‡]For $\{x, \bar{x}, \bar{x}\}$: if $T \neq 0$, then $\varepsilon_2 + \varepsilon_3 = 0$ forces $\varepsilon_1 = 0$; if $T = 0$, the x -equation gives $\varepsilon_1 = \varepsilon_2 + \varepsilon_3 \in \{0, \pm 2\}$, impossible since $\varepsilon_1 = \pm 1$.

[†]The case $\{x, \bar{x}, |x|^2\}$ arises only in the imaginary case ($\bar{x} \neq x$). The constraint $N = T$ gives $|N_{K/\mathbb{Q}}(x)| = \text{Tr}_{K/\mathbb{Q}}(x)$: for $d \equiv 1 \pmod{4}$, $(1-d)/4 = 1$ gives $d = -3$ (Row 3a via a different presentation). The *rational case* $x = 2$ ($K = \mathbb{Q}$) does not admit a $\mathbb{Q}$ -basis decomposition; it is verified directly: $1 + 1 - 2 = 0$ (Row 1). This accounts for all 20 unordered triples; every surviving solution maps to one of the seven rows of Table 1, completing the enumeration.

Row 3b merits comment: the sign pattern $(\varepsilon_1, \varepsilon_2, \varepsilon_3) = (1, -1, -1)$ on the triple $\{1, x, \bar{x}\}$ gives $1 - x - \bar{x} = 1 - T = 0$, hence $T = 1$ with no constraint on N . This vanishing sum holds for all imaginary quadratic fields with $d \equiv 1 \pmod{4}$ (e.g., $d = -11, -15, -19, \dots$). Unlike Row 3a ($T = -1$, which produces the Eisenstein island at $d = -3$), Row 3b does not lead to KS-uncolorability: the all-nonzero triads it generates are structurally decoupled from the one-zero triads, and the full ray set remains colorable. This is proved for all $N \geq 3$ by SMT verification [9]; see Lemma 2 for the proof.

Row 4 merits comment: the constraint $|x|^2 = 1 + x$ is satisfied by $x = \varphi = \frac{1+\sqrt{5}}{2}$ (golden ratio), which has $N_{K/\mathbb{Q}}(\varphi) = -1$ and $\text{Tr}_{K/\mathbb{Q}}(\varphi) = 1$. This does not satisfy (i) or (ii) of Theorem 2. Although $\mathbb{Q}(\varphi)$ -alphabet KS sets exist [3] with 52 vectors, they require cross-product completion beyond the initial alphabet $\mathcal{A}$ and thus lie outside the scope of the present classification. $\square$

Lemma 2 (Colorability of the complement). *Suppose x satisfies none of conditions (R), (i), or (ii) of Theorem 2. Then the ray set $S(\mathcal{A})$ admits an explicit valid $\{0, 1\}$ -coloring.*

Strategy. The proof splits the residual case into two subfamilies according to whether the Row-2 vanishing sum $1 + 1 - |x|^2 = 0$ is available.

Subfamily A (*no Row-2 cancellation*): $|x|^2 \neq 2$. Then by Lemma 1 the only surviving vanishing sum is Row 3b ($\text{Tr}_{K/\mathbb{Q}}(x) = 1$, available when $d \equiv 1 \pmod{4}$ imaginary), and Row 4 (golden ratio, $x = \varphi$, $|x|^2 = 1 + x$) which falls outside $\mathcal{O}_K$ for any $d \neq 5$ and is treated separately in Section 5. We show that the all-nonzero triads from Row 3b are structurally decoupled from the one-zero triads, and no type-B triad (mixing the two families) exists. The decoupling reduces colorability to two independent subproblems, both of which are explicitly colorable.

Subfamily B (*Row-2 cancellation, but $\text{Tr}_{K/\mathbb{Q}}(x) \neq 0$*): $|x|^2 = 2$ and $\text{Tr}_{K/\mathbb{Q}}(x) \neq 0$. This is the *non-pure* modulus-2 case, where condition (i) of Theorem 2 fails (it requires $\text{Tr}_{K/\mathbb{Q}}(x) = 0$ as well). Solving $|a + bx|^2 = 2$ over $\mathcal{O}_K$ with $\text{Tr}_{K/\mathbb{Q}}(x) \neq 0$, the alphabet is—up to sign and Galois-conjugation—one of exactly two:

- the Gaussian alphabet $\{0, \pm 1, \pm(1+i)\}$ over $K = \mathbb{Q}(i)$ (standard generator, $\text{Tr}_{K/\mathbb{Q}}(x) = 2$);
- the Heegner-7 halfint alphabet $\{0, \pm 1, \pm(1+\sqrt{-7})/2\}$ over $K = \mathbb{Q}(\sqrt{-7})$ ($\text{Tr}_{K/\mathbb{Q}}(x) = 1$).

For each of these two alphabets we exhibit an explicit valid $\{0, 1\}$ -coloring (Table 2), verified both by direct constraint check and by an independent SAT solver (Glucose4, PySAT [?]). These two cases exhaust subfamily B; no further analytical argument is needed. The coexistence of type-B triads (which exist here, by the Row-2 sum) with valid colorings shows that Row-2 cancellation alone is not a sufficient KS-uncolorability mechanism; the additional constraint $\text{Tr}_{K/\mathbb{Q}}(x) = 0$ —as in condition (i)—is required to pack enough triads to force unsatisfiability.

Table 2: Explicit valid $\{0, 1\}$ -colorings (subfamily B). Each row lists the green rays ($f(v) = 1$) of a coloring of the 49-ray pool of the corresponding alphabet (the remaining rays are red, $f(v) = 0$). Both colorings have been verified by direct constraint check (78 Hermitian-orthogonal pair constraints, 14 triad-completeness constraints) and independently by SAT (Glucose4, PySAT [?]); reproducer scripts are included with this paper.

Alphabet	Green rays (representative vectors in $\mathcal{A}^3$)
Gaussian $x = 1 + i$, $K = \mathbb{Q}(i)$	$(1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1+i), (1, 1+i, 1), (1, 1+i, 1+i),$ $(1, 1+i, -1-i), (1, -1-i, 1+i), (1, -1-i, -1-i), (1+i, 1, 1),$ $(1+i, 1, -1)$
Heegner-7 halfint $x = (1+\sqrt{-7})/2$, $K = \mathbb{Q}(\sqrt{-7})$	$(0, 0, 1), (0, 1, 1), (1, 0, 1), (1, 1, x), (1, -1, -x), (1, -x, 1), (x, 1, 1)$

Proof of Lemma 2. Subfamily B ($|x|^2 = 2$, $\text{Tr}_{K/\mathbb{Q}}(x) \neq 0$). By the count above, the alphabet is one of two specific cases. Table 2 exhibits an explicit valid $\{0, 1\}$ -coloring for each, completing this subfamily. We turn to subfamily A.

Subfamily A ($|x|^2 \neq 2$). By Lemma 1, Rows 1–3a and 5–6 are excluded by the negation of (R), (i), and (ii). Row 4 ($x = \varphi$) is treated in Section 5. For the remaining generators, the only surviving vanishing sum is Row 3b: $1 - x - \bar{x} = 0$, which holds precisely when $\text{Tr}_{K/\mathbb{Q}}(x) = 1$ (imaginary quadratic, $d \equiv 1 \pmod{4}$). When $\text{Tr}_{K/\mathbb{Q}}(x) \neq 1$, no vanishing sum exists at all. The remaining proof proceeds in three steps.

Step 1 (All-nonzero triads are harmless). When $\text{Tr}_{K/\mathbb{Q}}(x) \neq 1$ (including all real quadratic and all imaginary quadratic with $d \equiv 2, 3 \pmod{4}$), no primitive three-term vanishing sum from $\bar{\mathcal{A}} \cdot \mathcal{A}$ exists, so no two all-nonzero rays in $S(\mathcal{A})$ are orthogonal.

When $\text{Tr}_{K/\mathbb{Q}}(x) = 1$, the vanishing sum $1 - x - \bar{x} = 0$ does produce all-nonzero orthogonal pairs and, indeed, complete all-nonzero triads. Concretely, the triad $\{(1, 1, x), (1, -x, -1), (x, -1, 1)\}$ is mutually orthogonal (each inner product reduces to $1 - x - \bar{x} = 0$). However, these triads involve *only* all-nonzero rays and share no rays with one-zero triads. That no type-B triad (mixing all-nonzero and one-zero rays) exists is proved universally: the Hermitian cross product of any orthogonal (all-nonzero, one-zero) pair produces a vector whose $\mathbb{Q}$ -basis components cannot simultaneously lie in $\mathcal{A}$ for any integer $N \geq 3$, certified by the Z3 SMT solver [9] over all 1024 entry patterns and all integer (N, T) with $N \geq 3$.

Since the all-nonzero and one-zero families share no rays, a $\{0, 1\}$ -coloring is valid if and only if it is valid on each family separately. Both families are independently colorable (Step 3 below gives 192 colorings for the one-zero family; the all-nonzero triads, being pairwise disjoint, are trivially colorable). The all-nonzero triads are therefore *harmless*—they do not contribute to KS-uncolorability.

Step 2 (No type-B triads). Call a triad *type-B* if it contains at least one all-nonzero ray alongside rays with a zero coordinate. An all-nonzero ray $v \in \{\pm 1, \pm x\}^3$ can be orthogonal to a one-zero ray w (say $w_3 = 0$) via two-term cancellation: $\bar{v}_1 w_1 + \bar{v}_2 w_2 = 0$. However, the third member of any triad containing v and w is the unique ray in the Hermitian orthogonal complement of $\text{span}(v, w)$. In $\mathbb{C}^3$, this ray is represented by the

conjugate of the standard cross product,

$$u_k = \overline{v_{k+1}w_{k+2} - v_{k+2}w_{k+1}} = \bar{v}_{k+1}\bar{w}_{k+2} - \bar{v}_{k+2}\bar{w}_{k+1} \quad (\text{indices mod } 3),$$

which satisfies $\langle u|v \rangle = \langle u|w \rangle = 0$ for the Hermitian product $\langle a|b \rangle = \sum_k \bar{a}_k b_k$ because $\langle u|v \rangle = \sum_k (v \times w)_k v_k = \det[v, v, w] = 0$ and similarly $\langle u|w \rangle = \det[v, w, w] = 0$. (For real fields, $\bar{v}_k = v_k$ and u reduces to the standard bilinear cross product.) We show that $u \notin \mathcal{A}^3$ up to projective equivalence.

With $w_3 = 0$, the (conjugated) cross product gives $u = (-\bar{v}_3\bar{w}_2, \bar{v}_3\bar{w}_1, \bar{v}_1\bar{w}_2 - \bar{v}_2\bar{w}_1)$. Writing $\bar{w}_k \in \{\pm 1, \pm \bar{x}\}$ and $\bar{v}_k \in \{\pm 1, \pm \bar{x}\}$, the first two components lie in $\bar{\mathcal{A}} \cdot \bar{\mathcal{A}}$ and the third is a difference of two such products. For real quadratic ($\bar{x} = x$) this recovers $(-v_3w_2, v_3w_1, v_1w_2 - v_2w_1)$ as before. For imaginary quadratic with $d \equiv 2, 3 \pmod{4}$ (standard generator, $\bar{x} = -x$), the conjugate alphabet satisfies $\bar{\mathcal{A}} = \mathcal{A}$ as a set and the analysis proceeds by direct substitution. For $d \equiv 1 \pmod{4}$ (canonical halfint generator, $\bar{x} = 1 - x \neq \pm x$), the components of u lie a priori in $\bar{\mathcal{A}}^3$, and the test whether $[u]$ is projectively equivalent to a ray in $S(\mathcal{A})$ requires comparing u to some $\lambda w'$ with $w' \in \mathcal{A}^3$ and $\lambda \in K^\times$; since K is closed under conjugation we may equivalently test the conjugate configuration $\bar{u} = v \times w$ against $\bar{\lambda}\bar{\mathcal{A}}^3$, which is the original cross-product analysis over the conjugate alphabet. We will present the calculation uniformly, flagging the $d \equiv 1 \pmod{4}$ specializations where they diverge.

Real case ($d > 0$, so $\bar{v}_k = v_k$). The cross-product components are $u_1 = -v_3w_2$, $u_2 = v_3w_1$, $u_3 = v_1w_2 - v_2w_1$, with $v_k \in \{\pm 1, \pm x\}$ and $w_k \in \{\pm 1, \pm x\}$. For $u \in \lambda\mathcal{A}^3$, all three components must simultaneously lie in $\{0, \pm\lambda, \pm\lambda x\}$.

The first two components determine λ . Since $u_1 = -v_3w_2$ and $u_2 = v_3w_1$ are both nonzero (products of nonzero alphabet entries), each is $\pm\lambda$ or $\pm\lambda x$, giving four cases for the pair (u_1, u_2) :

$$\begin{aligned} u_1 = \pm\lambda, u_2 = \pm\lambda &\Rightarrow \lambda = \pm v_3w_2, u_2/u_1 = -w_1/w_2 = \pm 1 \\ u_1 = \pm\lambda, u_2 = \pm\lambda x &\Rightarrow u_2/u_1 = -w_1/w_2 = \pm x \\ u_1 = \pm\lambda x, u_2 = \pm\lambda &\Rightarrow u_2/u_1 = -w_1/w_2 = \pm 1/x \\ u_1 = \pm\lambda x, u_2 = \pm\lambda x &\Rightarrow u_2/u_1 = -w_1/w_2 = \pm 1 \end{aligned}$$

In every case, λ is determined by v_3 and one of w_k . The constraint then falls on u_3 : we need $u_3/\lambda \in \{0, \pm 1, \pm x\}$, equivalently $u_3/u_1 \in \{0, \pm 1, \pm x, \pm 1/x\}$.

The third component $u_3 = v_1w_2 - v_2w_1$ evaluates to one of $0, \pm 2, \pm 2x, \pm(x^2 \pm 1)$, or $\pm(x \pm 1)$. We check each against the constraint $u_3/u_1 \in \{0, \pm 1, \pm x, \pm 1/x\}$, where $u_1 \in \{\pm 1, \pm x, \pm x^2\}$:

- $u_3 = \pm 2$: $u_3/u_1 = \pm 2/u_1$. For $u_1 = \pm 1$: ratio $= \pm 2 \notin \{0, \pm 1, \pm x, \pm 1/x\}$ (since $x > 1, x \neq 2$ unless condition (i)). For $u_1 = \pm x$: ratio $= \pm 2/x$; need $2/x = 1$ ($x = 2$, condition (i)) or $2/x = x$ ($x^2 = 2$, condition (i)). For $u_1 = \pm x^2$: $2/x^2 \notin \{1, x, 1/x\}$ for $x > 1$.
- $u_3 = \pm 2x$: same scaling obstruction as ± 2 (ratio $\pm 2x/u_1$ yields ± 2 or $\pm 2x^2/1$, none in the target set for $x > 1, x \neq 2$).
- $u_3 = \pm(x^2 + 1)$: since $u_1 \in \{\pm 1, \pm x, \pm x^2\}$, the ratio $u_3/u_1 = \pm(x^2 + 1)/u_1$ must lie in $\{0, \pm 1, \pm x, \pm 1/x\}$. We check each u_1 : $u_1 = \pm 1$: ratio $= \pm(x^2 + 1) > 1$ and irrational for $x > 1$; not in target set. $u_1 = \pm x$: ratio $= \pm(x^2 + 1)/x = \pm(x + 1/x)$;

this equals ± 1 iff $x^2 - x + 1 = 0$ (no real root), and equals $\pm x$ iff $x^2 + 1 = x^2$ (false).
 $u_1 = \pm x^2$: ratio $= \pm(1 + 1/x^2)$; this equals ± 1 iff $x \rightarrow \infty$, and $\pm x$ iff $x^2 + 1 = x^3$ (no solution for $x > 1$ algebraic integer of degree 2).

- $u_3 = \pm(x^2 - 1)$: need $(x^2 - 1)/u_1 \in \text{target set}$. For $u_1 = x$: $(x^2 - 1)/x = x - 1/x$; this equals 1 iff $x^2 - x - 1 = 0$, i.e. $x = \varphi$ (excluded by hypothesis).
- $u_3 = \pm(x \pm 1)$: for $u_1 = \pm 1$: need $x \pm 1 \in \{0, 1, x, 1/x\}$, giving $x \in \{0, \pm 1, 2\}$ (all excluded or condition (i)). For $u_1 = \pm x$: need $(x \pm 1)/x = 1 \pm 1/x \in \{0, 1, x, 1/x\}$; $1/x$ equals $1/x$ iff the sign is $-$ (giving ratio $= 1 - 1/x$, which equals 1 only if $x \rightarrow \infty$). No solution for $x > 1$.

In every case, u_3/u_1 lies outside the target set unless x satisfies condition (i) or Row 4 (golden ratio, excluded by hypothesis).

Imaginary case ($d < 0$, so $\bar{v}_k \in \{\pm 1, \pm \bar{x}\}$ with $\bar{x} = \sigma(x) \neq x$). Since all components of u lie in K and the alphabet $\mathcal{A} \subset K$, any projective scalar $\lambda \in \mathbb{C}^\times$ satisfying $u = \lambda w$ with $w \in \mathcal{A}^3$ must itself lie in $K^\times$ (as $\lambda = u_k/w_k$ for any nonzero component). For $u \in \lambda \mathcal{A}^3$, each component must equal $\pm \lambda$ or $\pm \lambda x$.

We split into two sub-cases by the Galois trace.

Sub-case $\text{Tr}_{K/\mathbb{Q}}(x) \neq 0$ (i.e., $d \equiv 1 \pmod{4}$), $x = (1 + \sqrt{d})/2$, $\bar{x} = (1 - \sqrt{d})/2 = 1 - x$. The first two components are $u_1 = -\bar{v}_3 w_2$ and $u_2 = \bar{v}_3 w_1$, where $\bar{v}_3 \in \{\pm 1, \pm \bar{x}\} = \{\pm 1, \pm(1-x)\}$ and $w_k \in \{\pm 1, \pm x\}$. Writing $\bar{x} = 1 - x$ and using the minimal polynomial $x^2 = x + (d-1)/4$, we have $\bar{x} \cdot x = x - x^2 = -(d-1)/4 = -N'$. The ratio $u_1/u_2 = -w_2/w_1 \in \{\pm 1, \pm x, \pm x^{-1}\}$. For $u \in \lambda \mathcal{A}^3$ we need $u_1/u_2 \in \{\pm 1, \pm x, \pm x^{-1}\}$ (the ratios of alphabet elements), which is automatically satisfied. The constraint comes from the *third* component: $u_3 = \bar{v}_1 w_2 - \bar{v}_2 w_1$. Each term $\bar{v}_k w_j$ with $\bar{v}_k \in \{\pm 1, \pm(1-x)\}$ and $w_j \in \{\pm 1, \pm x\}$ takes values in $\{\pm 1, \pm x, \pm(1-x), \pm x(1-x)\}$. The difference u_3 therefore lies in $\{0, \pm 2, \pm 2x, \pm 2(1-x), \pm(1-2x), \pm(2x-1), \pm x(1-x) \pm 1, \dots\}$.

For u_3/λ to lie in $\{0, \pm 1, \pm x\}$ with λ determined by $u_1 = \pm \lambda$ or $\pm \lambda x$, we need $u_3/u_1 \in \{0, \pm 1, \pm x, \pm x^{-1}\}$. We check: $u_3/u_1 = (\bar{v}_1 w_2 - \bar{v}_2 w_1)/(-\bar{v}_3 w_2)$. When $\bar{v}_3 = 1$, $w_2 = 1$: $u_3/u_1 = -(\bar{v}_1 - \bar{v}_2 w_1)$. With $\bar{v}_1 = 1 - x$ and $\bar{v}_2 w_1 = 1$: $u_3/u_1 = -(1 - x - 1) = x$. This is in the allowed set. But with $\bar{v}_1 = 1$ and $\bar{v}_2 = 1 - x$, $w_1 = x$: $u_3/u_1 = -(1 - (1-x)x) = -(1 - x + x^2)$. Now $x^2 = x + (d-1)/4$ (from the minimal polynomial $x^2 - x - (d-1)/4 = 0$), giving $u_3/u_1 = -(1 - x + x + (d-1)/4) = -(1 + (d-1)/4) = -(d+3)/4$. For this to lie in $\{0, \pm 1, \pm x, \pm x^{-1}\}$, it must be rational (which it is: $-(d+3)/4 \in \mathbb{Q}$) and equal to 0, ± 1 , or $\pm x^{-1} = \pm 4/(d-1) \cdot (1-x) + \dots$ (which is irrational). So we need $(d+3)/4 \in \{0, 1\}$, giving $d = -3$ (already handled: this is the Eisenstein case, condition (ii)) or $d = 1$ (excluded: $|x|^2 = 1$).

We tabulate all sign/entry combinations for u_3/u_1 in the $\text{Tr}_{K/\mathbb{Q}}(x) \neq 0$ sub-case. With $\bar{v}_k \in \{\pm 1, \pm(1-x)\}$ and $w_k \in \{\pm 1, \pm x\}$, the products $\bar{v}_k w_j$ take values in

$$\{\pm 1, \pm x, \pm(1-x), \pm x(1-x)\}.$$

Using $x(1-x) = x - x^2 = -N'$ where $N' = (d-1)/4$ (from $x^2 = x + N'$), this simplifies to $\{\pm 1, \pm x, \pm(1-x), \pm N'\}$.

The ratio

$$\frac{u_3}{u_1} = \frac{-(\bar{v}_1 w_2 - \bar{v}_2 w_1)}{\bar{v}_3 w_2}$$

takes values of the form $(p + qx)/(r + sx)$ where p, q, r, s are determined by the entry choices. For this ratio to lie in the *target set* $\{0, \pm 1, \pm x, \pm x^{-1}\}$, the numerator must be a $\mathbb{Q}$ -multiple of the denominator (for $\{0, \pm 1\}$) or the numerator times x must be a $\mathbb{Q}$ -multiple of the denominator (for $\{\pm x, \pm x^{-1}\}$). Writing both in the $\mathbb{Q}$ -basis $\{1, x\}$, each constraint is a pair of rational equations in N' .

We now classify all entry patterns by the structure of w . Up to projective equivalence, the one-zero ray w (with $w_3 = 0$) has entries $w_1, w_2 \in \{\pm 1, \pm x\}$. We split by the type of w_2 (after normalizing $w_1 = 1$ when possible).

Case $w_2 = \pm x$ (or equivalently $w = (x, \pm 1, 0)$). The orthogonality condition $\bar{v}_1 w_1 + \bar{v}_2 w_2 = 0$ requires $\bar{v}_2 = -\bar{v}_1 w_1 / w_2$. Since $\bar{v}_k \in \{\pm 1, \pm(1-x)\}$ and $w_1/w_2 \in \{\pm 1/x, \pm x\}$, the product $\bar{v}_1 w_1 / w_2$ must also lie in $\{\pm 1, \pm(1-x)\}$. We check all four values of $\bar{v}_1$ with $w_1 = 1, w_2 = x$:

- $\bar{v}_1 = 1$: $\bar{v}_2 = -1/x = -(x-1)/N'$. For this to lie in $\{\pm 1, \pm(1-x)\}$: $(x-1)/N' = \pm 1$ gives $N' = \pm(x-1)$, but $N' \in \mathbb{Q}$ and $x-1 \notin \mathbb{Q}$, so no solution; $(x-1)/N' = \pm(1-x)$ gives $N' = \mp 1$.
- $\bar{v}_1 = -1$: $\bar{v}_2 = 1/x$; same analysis, $N' = \mp 1$.
- $\bar{v}_1 = 1-x$: $\bar{v}_2 = -(1-x)/x = N'/x$; (N'/x) lies in $\{\pm 1\}$ iff $N' = \pm x$ (irrational, impossible); in $\{\pm(1-x)\}$ iff $N' = \pm x(1-x) = \mp N'$, giving $N' = 0$ (excluded).
- $\bar{v}_1 = -(1-x)$: same as above with opposite sign.

Every case requires $|N'| = 1$, i.e. $N' = \pm 1$. For $w = (x, \pm 1, 0)$: orthogonality gives $\bar{v}_1 x + \bar{v}_2(\pm 1) = 0$, so $\bar{v}_2 = \mp \bar{v}_1 x$. Then $\bar{v}_1 x \in \{\pm x, \pm(1-x)x\} = \{\pm x, \mp N'\}$, and $\bar{v}_2$ must lie in $\{\pm 1, \pm(1-x)\}$: $\mp x \notin$ this set, and $\pm N' \in \{\pm 1, \pm(1-x)\}$ iff $|N'| = 1$. Hence $w_2 = \pm x$ is possible only when $N' = \pm 1$, i.e. $d = -3$ (condition (ii)) or $d = 5$ (real, not imaginary).

Case $w_2 = \pm 1$ (with $w_1 = 1$). Orthogonality gives $\bar{v}_1 = -\bar{v}_2 w_2 = \mp \bar{v}_2$, so $\bar{v}_1$ and $\bar{v}_2$ are negatives (up to sign of w_2). The compatible pairs $(\bar{v}_1, \bar{v}_2)$ are: $(\pm 1, \mp 1)$ and $(\pm(1-x), \mp(1-x))$. For each, the cross-product components are:

- $(\bar{v}_1, \bar{v}_2) = (1, -1)$: $u_3 = w_2 + w_1 = \pm 2, u_1 = -\bar{v}_3 w_2$.
- $(\bar{v}_1, \bar{v}_2) = (1-x, -(1-x))$: $u_3 = (1-x)(w_2 + w_1) = \pm 2(1-x), u_1 = -\bar{v}_3 w_2$.

With $\bar{v}_3 \in \{\pm 1, \pm(1-x)\}$ and $w_2 = \pm 1$, the ratio u_3/u_1 takes (up to sign) three distinct values:

$$\frac{u_3}{u_1} \in \left\{ \pm 2, \pm \frac{2x}{N'}, \pm(2-2x) \right\}$$

where $2x/N'$ arises from $2/(1-x) = -2x/N'$ via $(1-x)^{-1} = -x/N'$.

We check each against the target set $\{0, \pm 1, \pm x, \pm x^{-1}\}$:

- $\pm 2 = \pm x$ iff $x = 2$ (condition (i)). $\pm 2 = \pm 1$ is false. $\pm 2 = \pm x^{-1}$ gives $x = 1/2$, not an algebraic integer.
- $\pm 2x/N' = \pm x$ iff $N' = \pm 2$, i.e. $d = -7$ or $d = 9$ (the latter not squarefree). For $d = -7$: $2x/(-2) = -x \in \{-x\}$. But $d = -7$ has $|x|^2 = 2$ (condition (i), excluded by hypothesis). $\pm 2x/N' = \pm 1$ requires $(0, 2/N') = (\pm 1, 0)$: impossible.

- $\pm(2 - 2x) = \pm 1$ requires $(2, -2) = (\pm 1, 0)$: impossible. $\pm(2 - 2x) = \pm x$ requires $(2, -2) = (0, \pm 1)$: impossible. $\pm(2 - 2x) = \pm x^{-1} = \pm(x-1)/N'$: comparing $\{1, x\}$ -coefficients gives $N' = -1$, i.e. $d = -3$ (condition (ii), excluded).

Summary. For $d \equiv 1 \pmod{4}$ with $d < 0$: $w_2 = \pm x$ requires $|N'| = 1$ (only $d = -3$); $w_2 = \pm 1$ produces ratios in the target set only when $N' \in \{-1, -2\}$ (i.e., $d \in \{-3, -7\}$), both excluded by hypothesis. Hence for $|N'| \geq 3$ (equivalently $|d| \geq 13$), no ratio u_3/u_1 lies in $\{0, \pm 1, \pm x, \pm x^{-1}\}$.

The borderline case $d = -11$ ($N' = -3$) is already covered: the $w_2 = \pm x$ case requires $|N'| = 1$ (fails), and the $w_2 = \pm 1$ ratios $\pm 2, \pm 2x/(-3), \pm(2-2x)$ are not in the target set (confirmed by exhaustive computation of all $4^3 \times 2 = 128$ sign/entry combinations, each yielding a ratio outside $\{0, \pm 1, \pm x, \pm x^{-1}\}$).

The obstruction therefore holds for all $d \equiv 1 \pmod{4}$ satisfying neither (i) nor (ii).

Sub-case $\text{Tr}_{K/\mathbb{Q}}(x) = 0$ (i.e., $d \equiv 2, 3 \pmod{4}$, $x = \sqrt{d}$, $\bar{x} = -x$). Here $\bar{x} = -x \in \mathcal{A}$, so the Hermitian cross product reduces to the real-case structure with $x^2 = d < 0$. Then u_3 takes values $\pm 2, \pm 2x, \pm(d \pm 1)$, and $\pm(x \pm 1)$. Since $|d| \geq 3$ by hypothesis ($d = -1$ gives $|x|^2 = 1$; $d = -2$ gives condition (i)), the scaling obstruction from the real case applies: $|d \pm 1| \geq 2$ cannot equal 1 or $|x| = \sqrt{|d|}$, and $|x \pm 1|^2 = |d| + 1 \geq 4$ prevents rescaling into $\{|\lambda|, |\lambda x|\}$.

In every case, u cannot be rescaled into $\mathcal{A}^3$. Therefore no triad in $S(\mathcal{A})$ contains an all-nonzero ray.

Step 3 (Explicit coloring). By Step 2, no triad mixes all-nonzero and one-zero rays. By Step 1, when $\text{Tr}_{K/\mathbb{Q}}(x) = 1$ the all-nonzero triads are independently colorable. It remains to color the one-zero triads.

Every one-zero triad consists of rays lying in coordinate planes. The only triad of axis-aligned rays is $\{e_1, e_2, e_3\}$. Within each coordinate plane (say $v_3 = 0$), the one-zero rays from $\mathcal{A}^3$ form orthogonal pairs, and the only complete triad is obtained by appending the axis ray e_3 .

For generic x (satisfying neither (i) nor (ii)), each coordinate plane contributes 6 projectively distinct one-zero rays and 3 orthogonal pairs, giving $3 \times 3 + 1 = 10$ one-zero triads. The coloring decomposes as follows: choose which axis ray receives 1 (3 choices); in each of the remaining two coordinate planes, the orthogonal pairs can be independently $\{0, 1\}$ -colored (2^3 choices per plane). This yields $3 \times 2^3 \times 2^3 = 192$ valid colorings of the one-zero family.

Edge cases: when $|x|^2 = 1$ (Gaussian-type), certain rays become projectively equivalent, reducing to ≤ 7 triads; when $\text{Tr}_{K/\mathbb{Q}}(x) = 0$, additional projective identifications occur but the triad count remains 10. In all cases the decomposition into axis choice plus independent planar pair-matchings persists, and the coloring is valid.

When $\text{Tr}_{K/\mathbb{Q}}(x) = 1$, the all-nonzero triads (4 projectively distinct, on 28 rays) are colored independently; since no all-nonzero ray appears in a one-zero triad (Step 1), the two colorings compose freely. The absence of type-B triads is proved for all $N \geq 3$ by SMT verification [9], not merely for sampled discriminants. $\square$

Proof of Theorem 2. Sufficiency is established by the explicit constructions cited above. For necessity, suppose neither (i) nor (ii) holds and $x \neq \varphi$. By Lemma 1, Rows 1–3a

and 5–6 are excluded (they require (i) or (ii)), Row 4 requires $x = \varphi$, and the only potentially surviving vanishing sum is Row 3b ($\text{Tr}_{K/\mathbb{Q}}(x) = 1$). Lemma 2 then produces an explicit valid $\{0, 1\}$ -coloring regardless of whether Row 3b applies, since the all-nonzero triads it permits are harmless (Step 1). The remaining case $x = \varphi$ is addressed in the Discussion: the raw golden-ratio alphabet is KS-colorable [3], confirming that neither (i) nor (ii) correctly predicts colorability for this generator as well. $\square$

4 Consequences

Corollary 1 (Imaginary quadratic classification). *Among imaginary quadratic fields $K = \mathbb{Q}(\sqrt{-d})$ with $d > 0$ squarefree, there exists a ring-of-integers generator y such that the raw ray set $S(\{0, \pm 1, \pm y\})$ is KS-uncolorable if and only if $d \in \{2, 3\}$:*

- $d = 2$: $y = \sqrt{-2}$, via condition (i) ($\text{Tr}_{K/\mathbb{Q}}(y) = 0$, $|y|^2 = 2$); the $\mathbb{Z}[\sqrt{-2}]$ island.
- $d = 3$: $y = \omega = (-1 + \sqrt{-3})/2$, via condition (ii) ($N_{K/\mathbb{Q}}(y) = 1$, $\text{Tr}_{K/\mathbb{Q}}(y) = -1$); the Eisenstein island.

Proof. Every $\mathbb{Z}$ -algebra generator of $\mathcal{O}_K$ has the form $y = a + \varepsilon x_0$ with $a \in \mathbb{Z}$, $\varepsilon = \pm 1$, and x_0 the canonical generator: $\{1, x_0\}$ is a $\mathbb{Z}$ -basis, and $\mathbb{Z}[y] = \mathcal{O}_K$ forces the x_0 -coefficient to be ± 1 . Since the nontrivial automorphism of K is complex conjugation, the Galois trace and norm are $\text{Tr}_{K/\mathbb{Q}}(y) = 2a + \varepsilon \text{Tr}_{K/\mathbb{Q}}(x_0)$ and $N_{K/\mathbb{Q}}(y) = |y|^2 = a^2 + \varepsilon a \text{Tr}_{K/\mathbb{Q}}(x_0) + |x_0|^2$. We split on $d \bmod 4$.

$d \equiv 1, 2 \pmod{4}$: here $x_0 = \sqrt{-d}$ with $\text{Tr}_{K/\mathbb{Q}}(x_0) = 0$, so $\text{Tr}_{K/\mathbb{Q}}(y) = 2a$ is even. Condition (ii) requires $|\text{Tr}_{K/\mathbb{Q}}(y)| = 1$, impossible. Condition (i) requires $\text{Tr}_{K/\mathbb{Q}}(y) = 0$, i.e. $a = 0$, whence $|y|^2 = d$; this equals 2 only for $d = 2$.

$d \equiv 3 \pmod{4}$: here $x_0 = (1 + \sqrt{-d})/2$ with $\text{Tr}_{K/\mathbb{Q}}(x_0) = 1$, so $\text{Tr}_{K/\mathbb{Q}}(y) = 2a + \varepsilon$ is odd. Condition (i) requires $\text{Tr}_{K/\mathbb{Q}}(y) = 0$, impossible. Condition (ii) requires $|\text{Tr}_{K/\mathbb{Q}}(y)| = 1$ (so $2a + \varepsilon = \pm 1$, i.e. $a \in \{0, -\varepsilon\}$) and $N_{K/\mathbb{Q}}(y) = 1$. Writing $N_{K/\mathbb{Q}}(y) = a^2 + \varepsilon a + (1+d)/4$, the constraint $N_{K/\mathbb{Q}}(y) = 1$ gives $d = 3 - 4a(a + \varepsilon)$; since $a \in \{0, -\varepsilon\}$ both yield $a(a + \varepsilon) = 0$, this forces $d = 3$.

Hence $d \in \{2, 3\}$, realized by $y = \sqrt{-2}$ and $y = \omega$ (equivalently $\bar{\omega}$) respectively. A SAT cross-check over all generators with $1 \leq d \leq 43$ confirms the raw ray set is uncolorable for exactly these two fields. $\square$

Remark 2 (The Heegner subset is a completed-pool phenomenon). *The four smallest Heegner numbers $d \in \{1, 2, 3, 7\}$ —the Gaussian $(1+i)$, $\sqrt{-2}$, Eisenstein (ω) , and Heegner-7 $((1+\sqrt{-7})/2)$ alphabets—each yield a KS set, but only two of them ($d = 2, 3$) do so at the raw level. The Gaussian and Heegner-7 alphabets have $|x|^2 = 2$ yet $\text{Tr}_{K/\mathbb{Q}}(x) \neq 0$, so they fail condition (i); their raw ray pools lack the triad density to force uncolorability (Lemma 2) and become KS sets only after cross-product completion (Remark 1). The Heegner-number characterization therefore belongs to the completed six-island picture [3], not to the raw classification of Theorem 1.*

Corollary 2 (Galois symmetry). *Every $\sigma \in \text{Gal}(K/\mathbb{Q})$ acts as a graph automorphism of the orthogonality graph and basis hypergraph of $S(\mathcal{A})$.*

Proof. The automorphism σ acts coordinate-wise on K^3 . For real quadratic K : complex conjugation is trivial ($\bar{v}_k = v_k$), so the Hermitian inner product reduces to the real dot product, and σ preserves it because σ is a ring homomorphism: $\sigma(\langle v|w \rangle) = \sigma(\sum v_k w_k) = \sum \sigma(v_k) \sigma(w_k) = \langle \sigma(v) | \sigma(w) \rangle$. For imaginary quadratic K : the non-trivial automorphism $\sigma(\sqrt{-d}) = -\sqrt{-d}$ coincides with complex conjugation restricted to K (since $\overline{\sqrt{-d}} = -\sqrt{-d}$), so $\sigma(\bar{v}_k) = \overline{\sigma(v_k)}$ and the Hermitian inner product is preserved. In both cases, σ permutes rays and maps triads to triads. $\square$

Connection to $\mathbb{Z}[1/6]$.—Conditions (i) and (ii) involve exactly the primes 2 and 3, matching the Cortez–Morales–Reyes result [7] that $M_3(\mathbb{Z}[1/N])_{\text{sym}}$ has no algebraic hidden states if and only if $6 \mid N$.

5 Discussion

The golden-ratio exception.—The golden ratio $x = \varphi = (1+\sqrt{5})/2$ is the unique quadratic generator satisfying neither condition (i) nor (ii) for which a Row 4 vanishing sum ($1+x-x^2=0$) exists. The analytical proof of Lemma 2 excludes this case because all-nonzero orthogonal pairs *do* arise. However, direct enumeration of $S(\mathcal{A})$ for $x = \varphi$ confirms that the raw alphabet admits 192 valid $\{0, 1\}$ -colorings [3], consistent with the theorem’s prediction. After cross-product completion, the $\mathbb{Q}(\varphi)$ -alphabet becomes KS-uncolorable (52 vectors [3]), showing that the raw/completed distinction is essential.

Scope and extensions.—The classification covers *raw* alphabets $\{0, \pm 1, \pm x\}$ only. Extension to cubic and higher-degree fields, and to completion-expanded algebras, remains open.

Logical chain.—Gleason [8] $\rightarrow$ Kochen–Specker [1] $\rightarrow$ algebraic islands [3] $\rightarrow$ this work: the Galois norm and trace classify which coordinate arithmetics support KS witnesses. Both the matrix trace (Gleason) and the Galois trace (Theorem 2) encode the primes 2 and 3. Whether a unified invariant from class field theory (perhaps the conductor of $\mathbb{Q}(\zeta_6)$, which is $6 = 2 \times 3$) simultaneously captures both conditions and generalizes to higher degree is the natural open question.

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